

THE EMERGENT BELT

涌现之带

Jing-Zhang Hyper Line

Centennial Jing-Zhang AI Belt · open-call submission

KUN-SAL Spatial Quant Lab · Xing Xinhua



$\lambda_c=0.42$

connection

$H=1.556$

health

$D=1.746$

fractal

25,476

POIs

+15.9%

job reach

44亿

investment

Conceptual: provisional boundary dashed only; statutory indicators pending

01

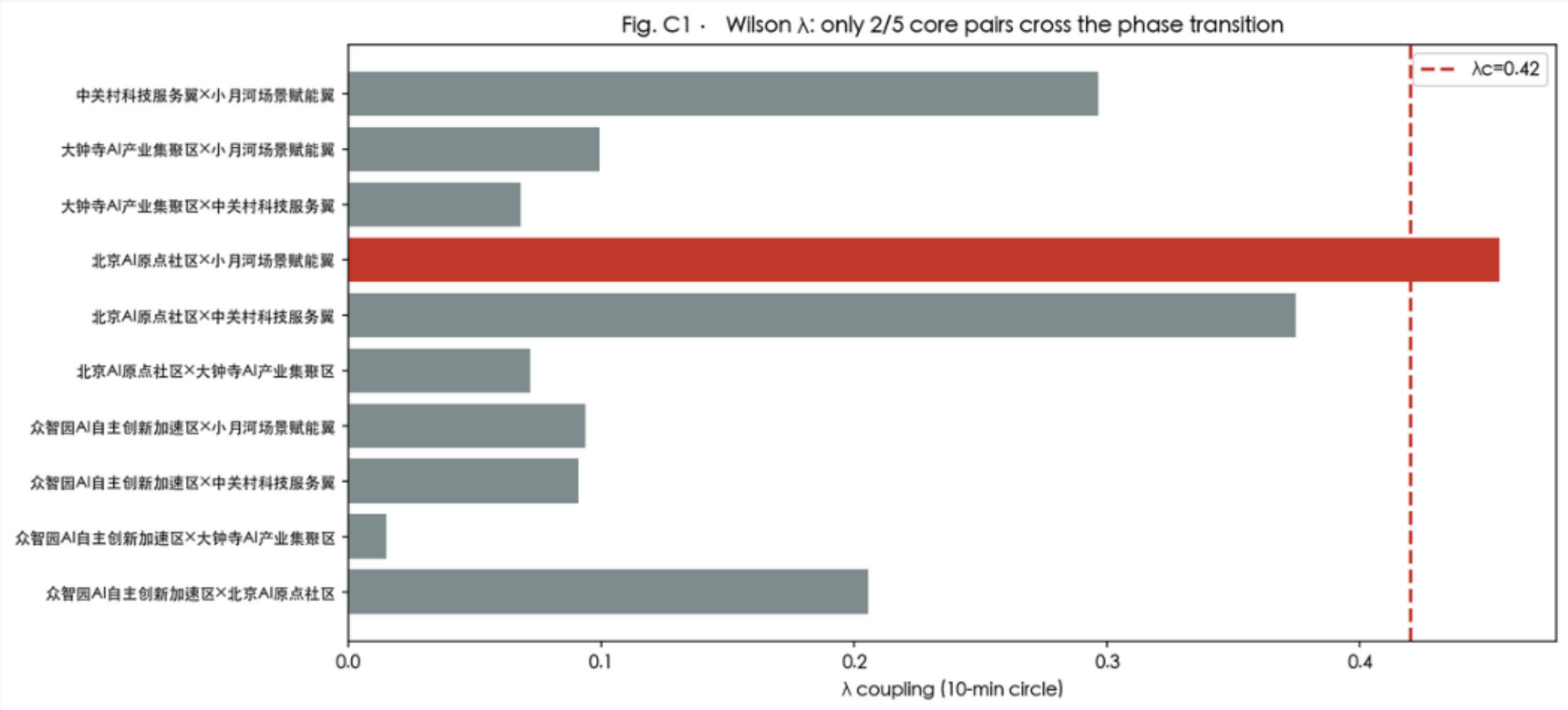
Diagnosis: what blocks this belt

Five scientists' consultation · Space syntax · LUTI (jobs-housing dynamics) · POI dynamics · Boundary correction

① Spatial interaction λ: stuck subcritical

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- Only one of ten core pairs crosses the line (Origin × Xiaoyuehe 0.455)
- Zhongzhiyuan×Dazhongsi λ=0.033: adjacent on the map, unreachable in ten minutes
- Not without vitality — vitality chopped apart by seven roads
- The leverage point is the seven gaps; no even spreading



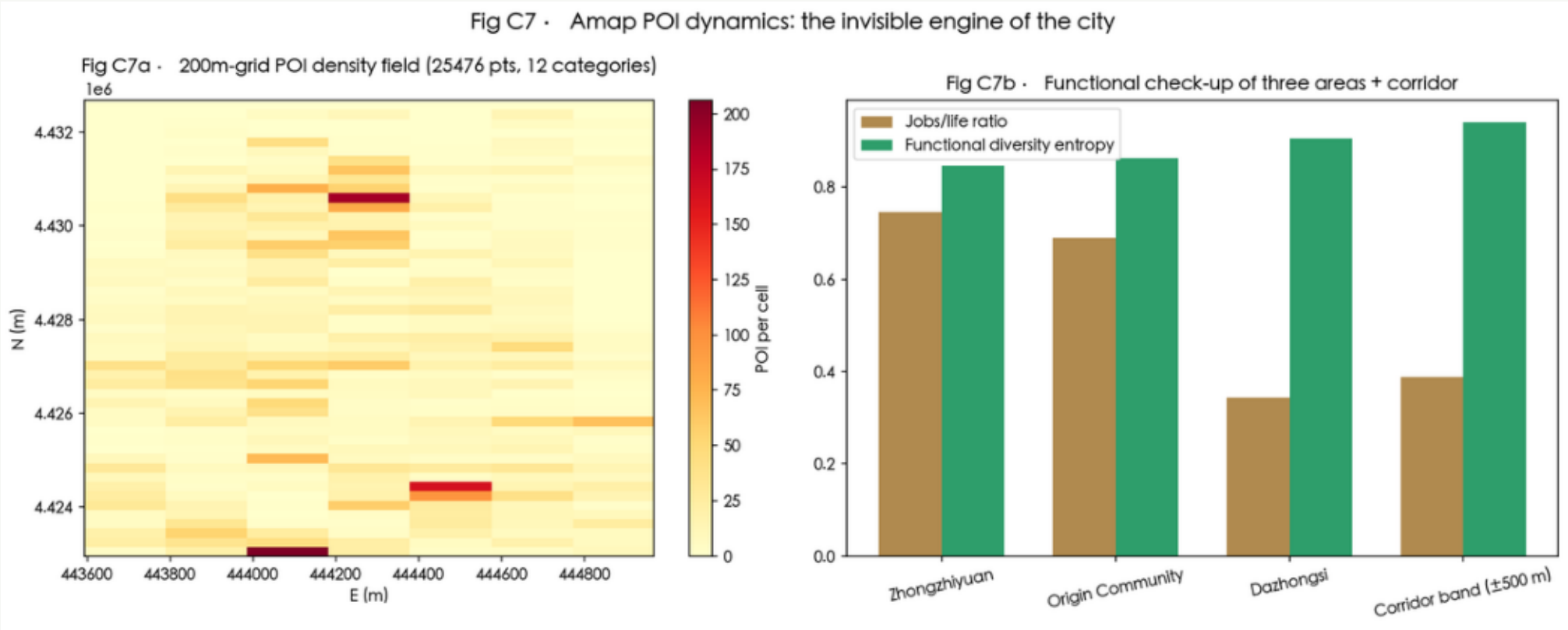
Basis: $\lambda = \exp(-d/2500m)$ proxy; recalculation on official data



② POI check-up: demand recognized by 25,476 points

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- 25,476 POIs across the belt (2026-08-15 snapshot)
- Life services over-supplied; innovation content under-supplied
- Finance –63.0% / research –45.9% / Origin medical 2.88/km², the lowest
- Dazhongsi 511/km², the densest; jobs/life 0.344: life dense, work thin

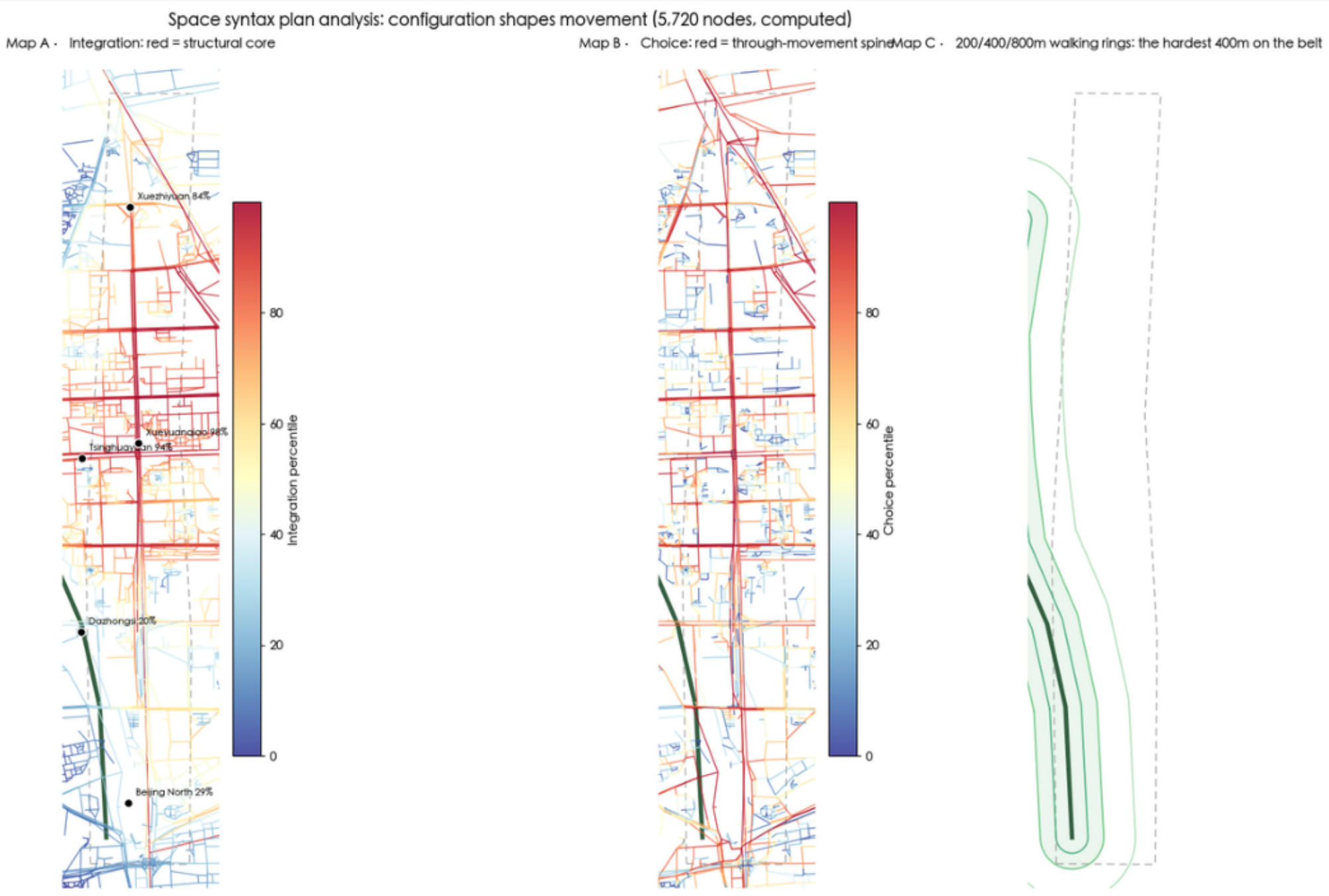


Data: Amap aggregate indicators; no raw points published



③ Space syntax: configuration shapes movement

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▪ Xueyuanqiao 97.5%: the belt's mouth at the N4th Ring × Xueyuanlu

▪ Tsinghuayuan 93.5%: a 1909 siting still the structural core

▪ Dazhongsi 20.2%: syntactic periphery, vitality must come from function

▪ After stitching: Dazhongsi +70%, Beijing North +49%, Wudaokou +43%

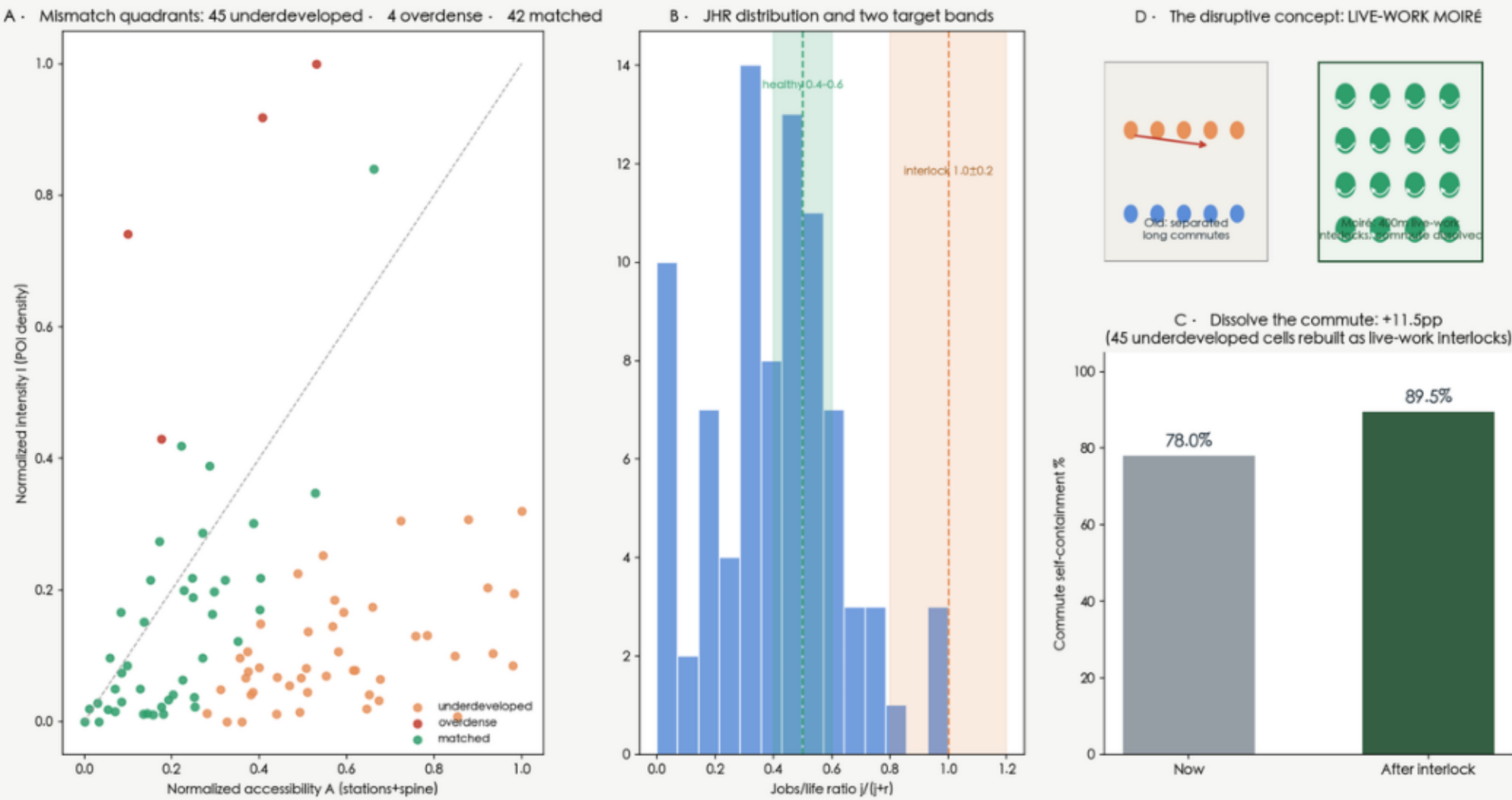
5,720-node network, computed; low intelligibility is normal for organic fabric



⑤ Deep LUTI: LIVE-WORK MOIRÉ — dissolve the commute

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Fig C27 · Deep LUTI analysis: LIVE-WORK MOIRÉ — from "shortening the commute" to "dissolving the commute"



Basis: POI proxies for jobs (companies+research+finance) and life side (residential+services+dining); the interlock is a conceptual inference — recalculation on official population and job data.

▪ Disruptive concept: 45 underdeveloped cells rebuilt as 1.0 ± 0.2 live-work interlocks

▪ Commute self-containment 78.0% → 89.5% (+11.5pp, computed)

▪ Mismatch quadrants 45/4/42: where it's wrong → after stitching → what remains

▪ Not a city with fewer commutes — a city that does not need commuting

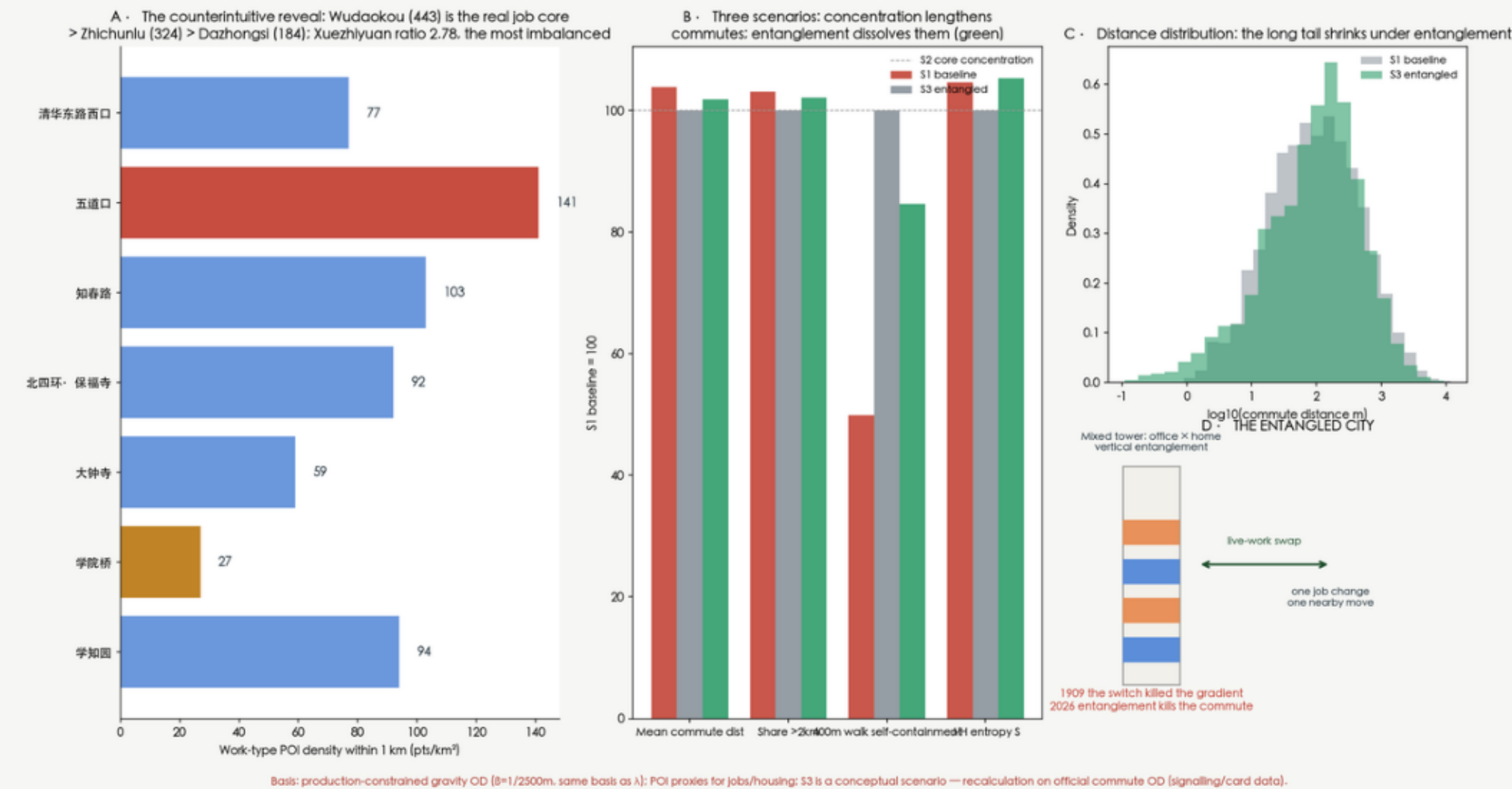
POI proxies: recalculation on official population and job data

⑥ Three-layer LUTI scenarios: concentration lengthens, entanglement dissolves

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▪ The reveal: Wudaokou 141/km² is the real job core; Xuezh yuan ratio 2.78, most imbalanced

Fig C28 · Three-layer LUTI scenario runs: only live-work entanglement dissolves the commute (Batty-style scenarios)



▪ Three scenarios (gravity OD): concentration +3.9% distance, walking self-containment 7.0%→3.5%

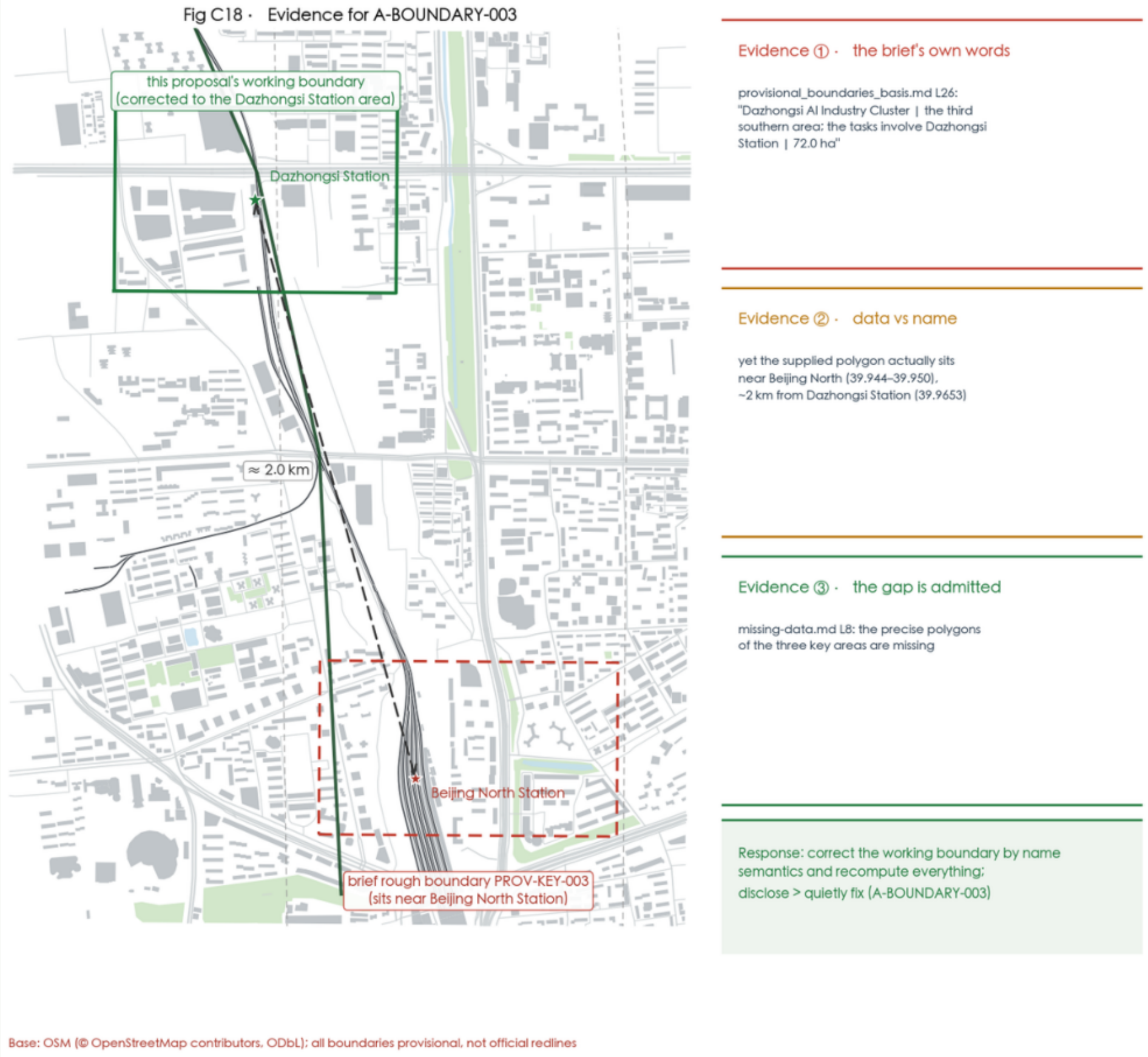
▪ Entangled: entropy S +5.3% (more mixed); rules = paired infill + mixed towers + S in the annual check-up

▪ 1909 the switch killed the gradient, 2026 entanglement kills the commuting tide

Gravity OD + POI proxies; recalculation on official OD; floorspace ledger = the brain's first new database

⑦ Boundary correction: fixing the brief's rough boundary

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- The brief's PROV-KEY-003 rough boundary sits near Beijing North (39.944–39.950)
- Its own basis table names Dazhongsi Station (39.9653) — ~2 km away
- missing-data.md: precise polygons are missing
- Corrected by name semantics, assumption A-BOUNDARY-003, full recalculation on official data

Integrity first: a data mismatch must be disclosed, never quietly fixed

02

Spatial proposal: one line, three folds, two wings, seven nodes

Master plan · Key-area plans & sections · Character & height · Dazhongsi quadrants

⑤ Master plan (real fabric + concept overlay)

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Fig.1 · Master Plan: one line, three folds, two wings, seven nodes (concept)



▪ Light grey = OSM building fabric; white-cased dark = real Jing-Zhang railway

▪ Dark green = Hyper Line spine; red dots = seven stitches; stars = three landmarks

▪ Three coloured frames = three gates: STACK / ORIGIN / FRONT

▪ The base is the real city; only the overlay is design

Base: OSM (© OpenStreetMap contributors)

⑥ Zhongzhiyuan · STACK (192.1 ha)

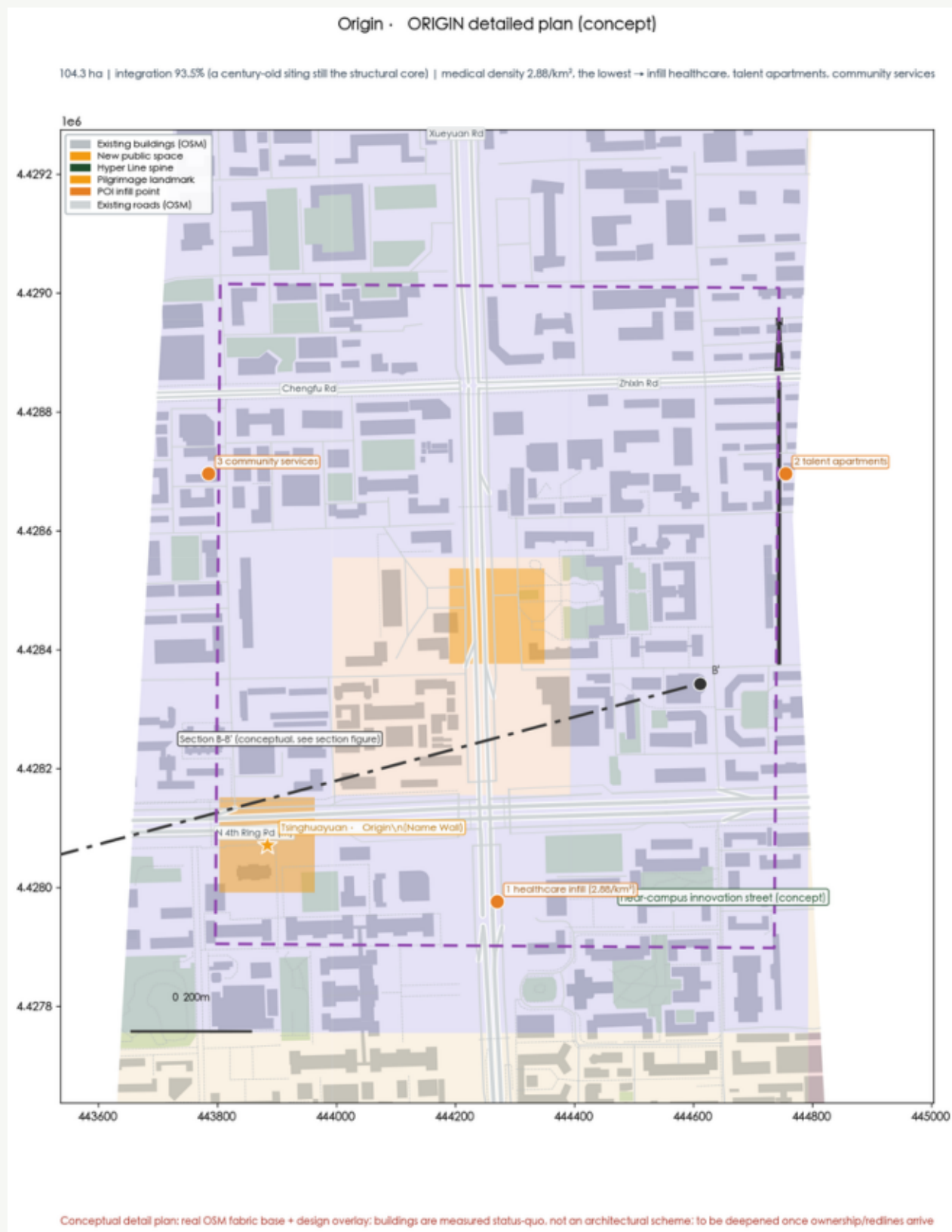
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- Qinghe full-stack test band: bounded tests in garden blocks
- Weakest: culture-tourism/education/finance → reasons to stay
- Section A-A': 3.2m footpath + 2.8m cycle band + test-equipment slots
- Long term (5–10y): garden-district renewal

⑦ Origin - ORIGIN (104.3 ha)

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- Tsinghuayuan · Origin Plaza + Name Wall: integration 93.5%
- Wudaokou · Phase Plaza: the λ gauge — the city's thermometer
- Medical 2.88/km², the lowest → infill healthcare/apartments/services
- Mid term (3–5y): near-campus innovation street

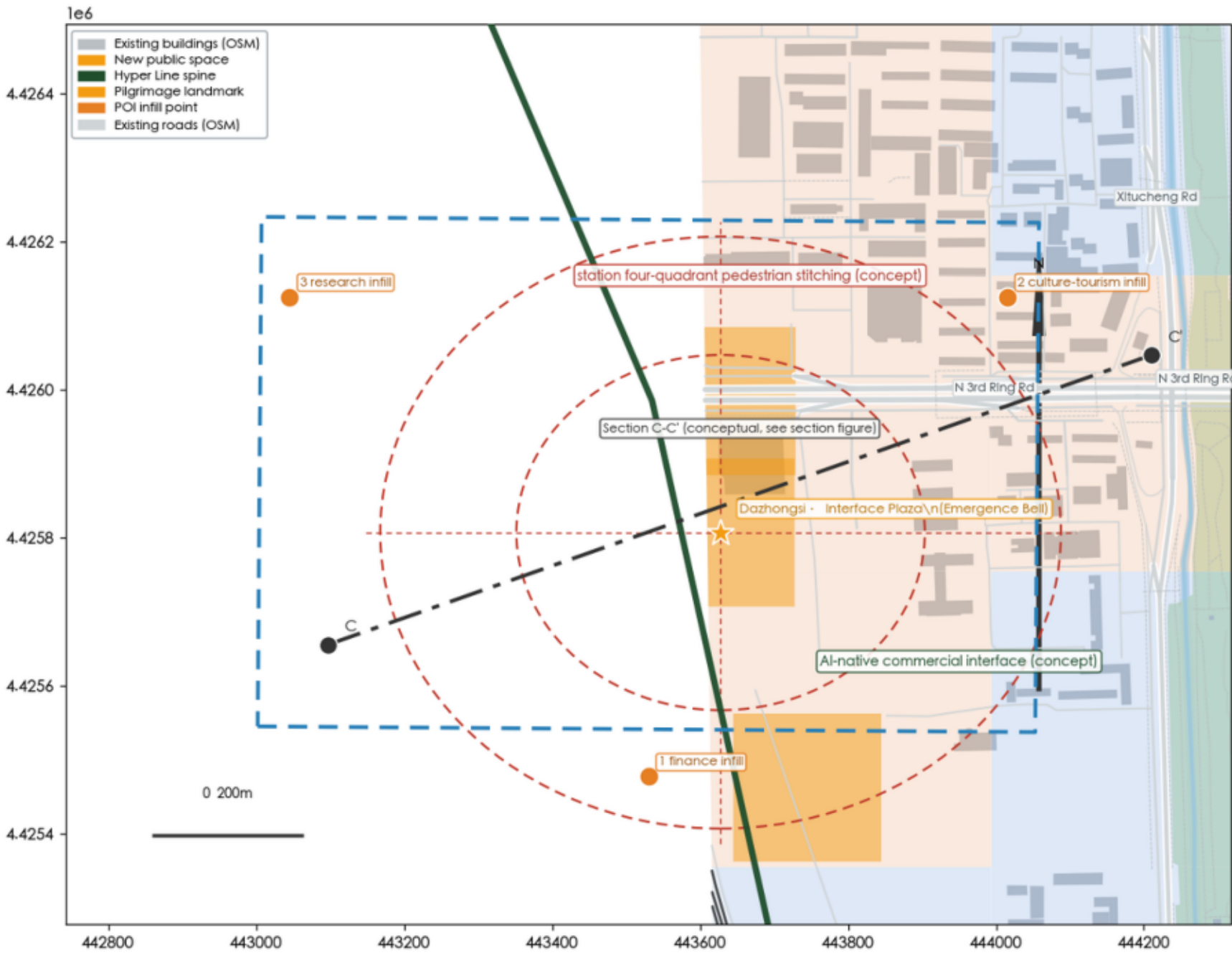
104.3 ha	93.5%	2.88/km ²
area of origin	structure of origin	mean of density

⑧ Dazhongsi · FRONT (72.0 ha)

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Dazhongsi · FRONT detailed plan (concept)

72.0 ha | syntactic percentile 20.2% (southern periphery) | POI 511/km², the densest of the three · jobs-residence 0.344 | infill finance, culture-tourism, research



Conceptual detail plan: real OSM fabric base + design overlay; buildings are measured status-quo, not an architectural scheme; to be deepened once ownership/redlines arrive

▪ Syntactic periphery 20.2%, yet the densest POIs at 511/km²

▪ Jobs/life 0.344: life-service-led → infill finance/culture-tourism/research

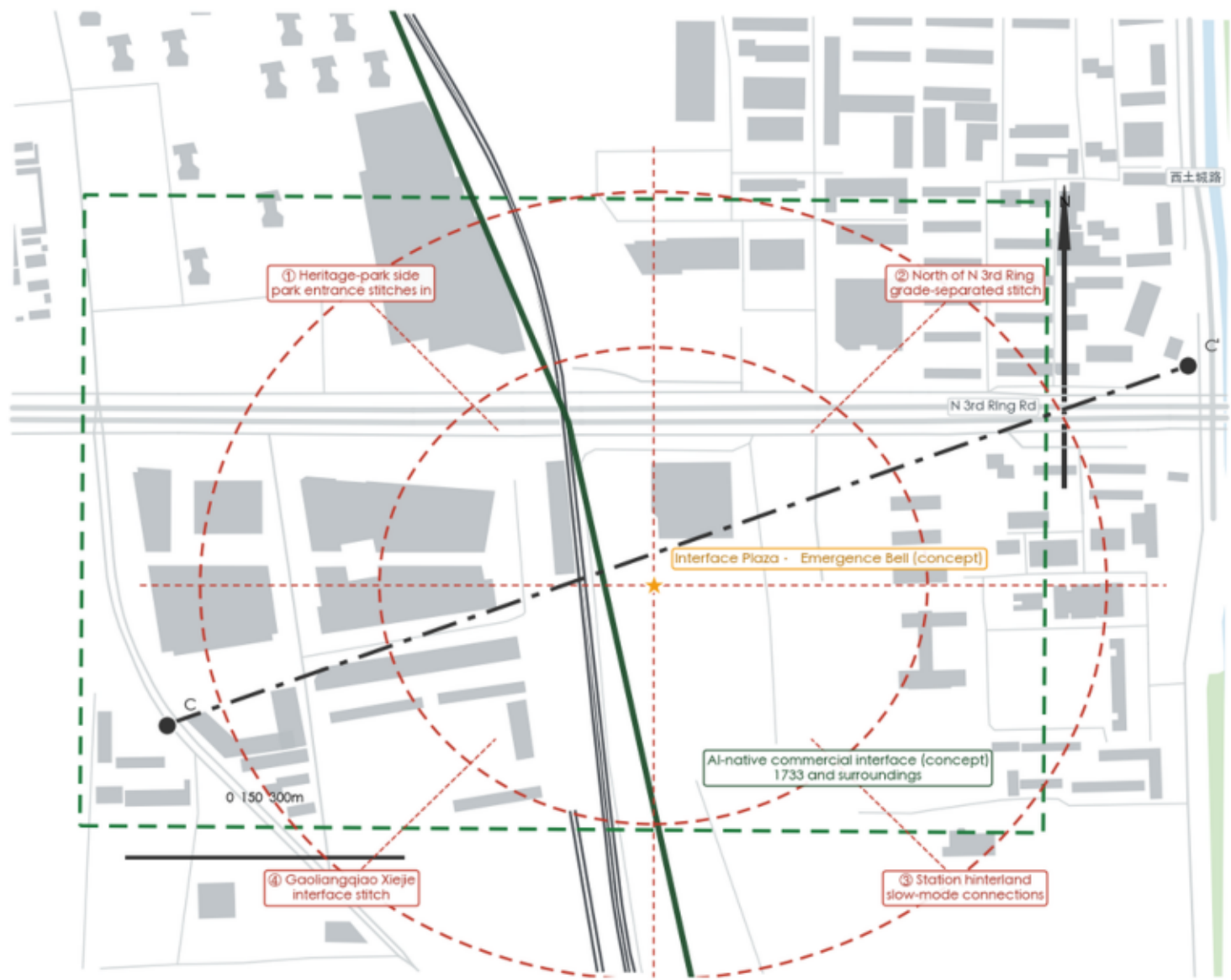
▪ Light intervention, heavy interface: stitch four quadrants; nights for classes & merchants

▪ Near term (1–3y): spine stitching + Dazhongsi interface

⑨ Dazhongsi four-quadrant stitching (enlarged)

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Fig C19 · Dazhongsi station-district four-quadrant stitching, enlarged (conceptual design)



Conceptual schematic: the stitch modes (grade-separation / slow-mode links) are conceptual suggestions, to be deepened once transport, municipal, and ownership data arrive; section C-C' shown in the detailed section figure.

Base: OSM (© OpenStreetMap contributors, ODbL)

- ① park entrance stitches in ② grade-separated stitch north of N3rd Ring
- ③ station-hinterland slow modes ④ Gaoliangqiao Xiejie interface
- Interface Plaza · Emergence Bell + the 1733 AI-native interface
- Section line C-C' matches the detailed plan

⑩ Street sections of the three areas (conceptual · dimension chain · plan index)

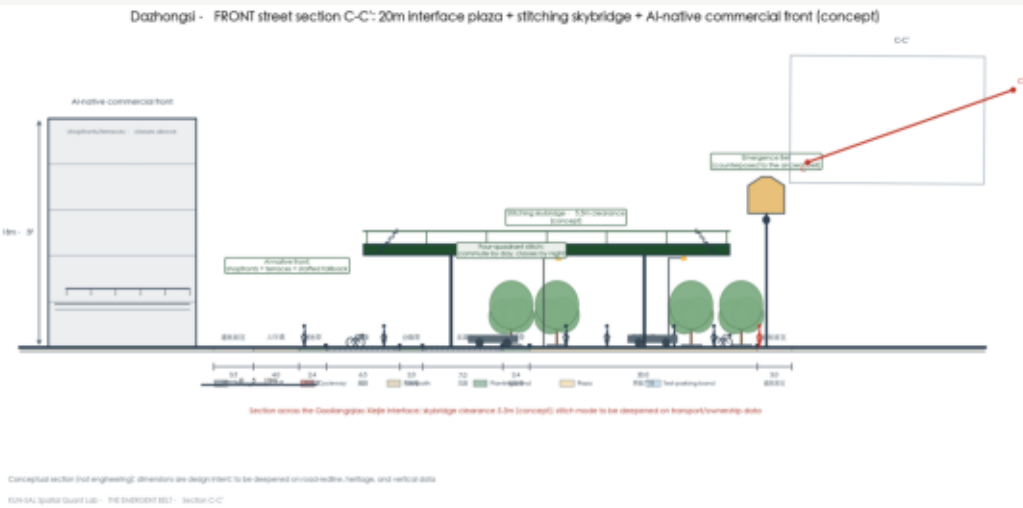
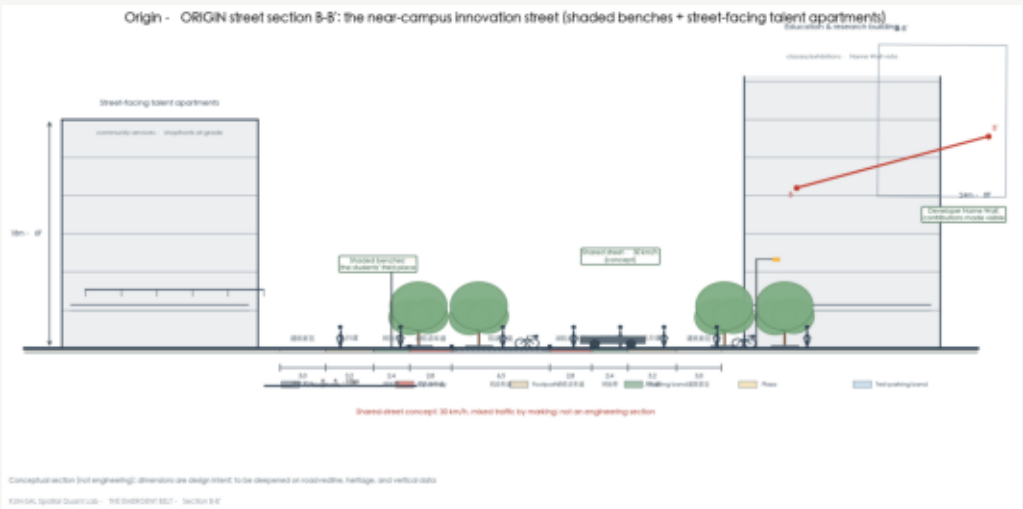
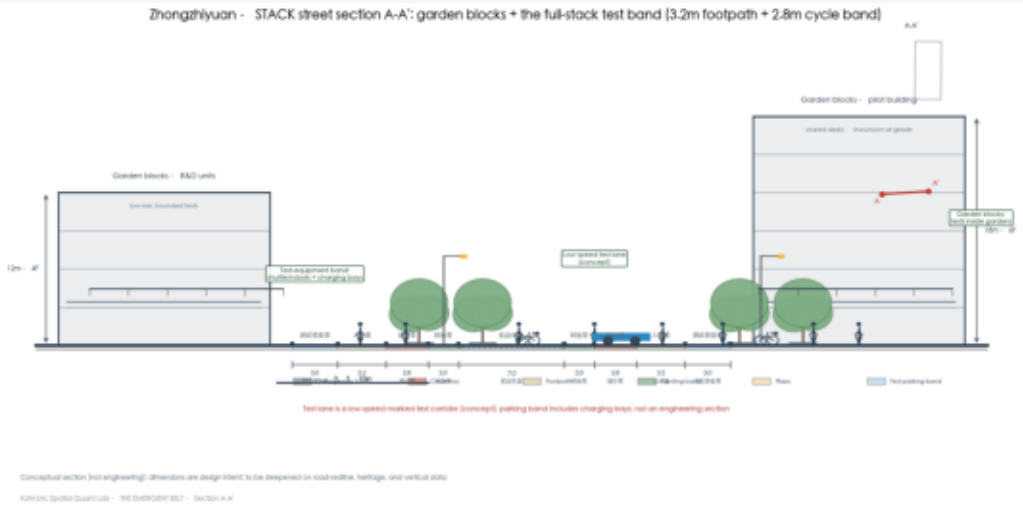
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▪ Each section: building-to-building full section + dimension chain + floors/heights

▪ Scale references: 1.8m people · bikes · cars · low-speed shuttle · trees & lamps

▪ Dazhongsi C-C': 20m interface plaza + stitching skybridge (5.5m) + Emergence Bell

▪ Conceptual sections (not engineering); dimensions pending redline & heritage data

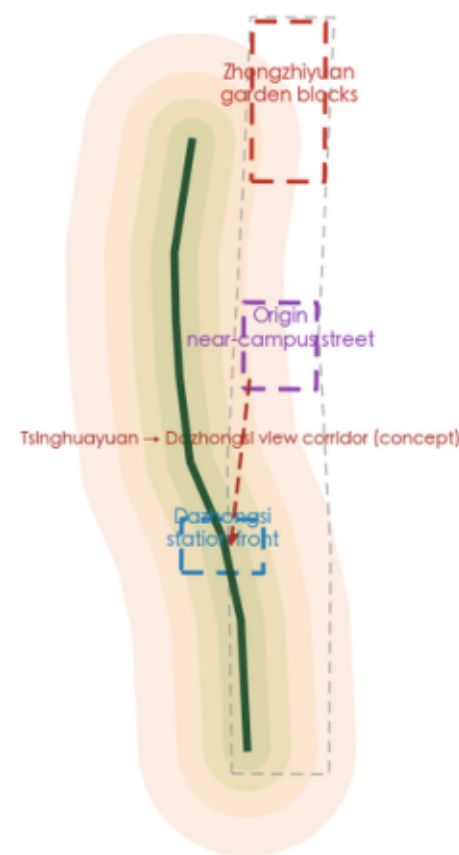


Conceptual sections: design intent; to be deepened on redline, heritage and vertical data

Character & conceptual height zoning

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Conceptual height zoning (schematic): bands step down toward the spine, low front / high back

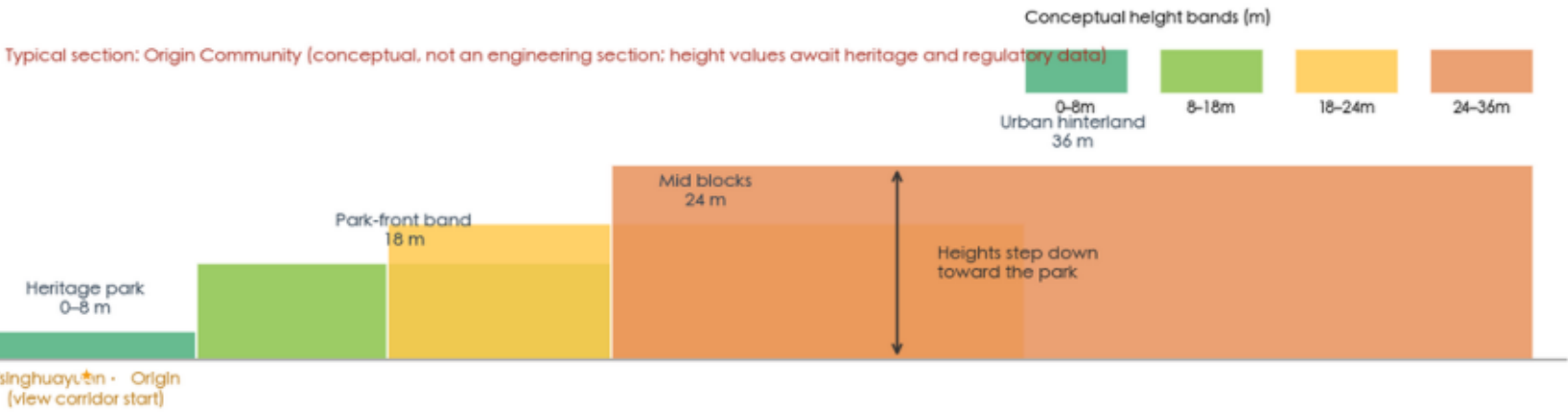


▪ Height bands step down along the spine: 0–8 / 18 / 24 / 36 m

▪ Three rules: garden blocks / near-campus street / station front

▪ Protective overlay (heritage + view corridors) + generative rules (low front, high back)

▪ Height values await heritage & regulatory data (not a regulatory drawing)



Schematic: height and character rules are design intent, to be rechecked once heritage, regulatory, and ownership data arrive (not a regulatory drawing)



03

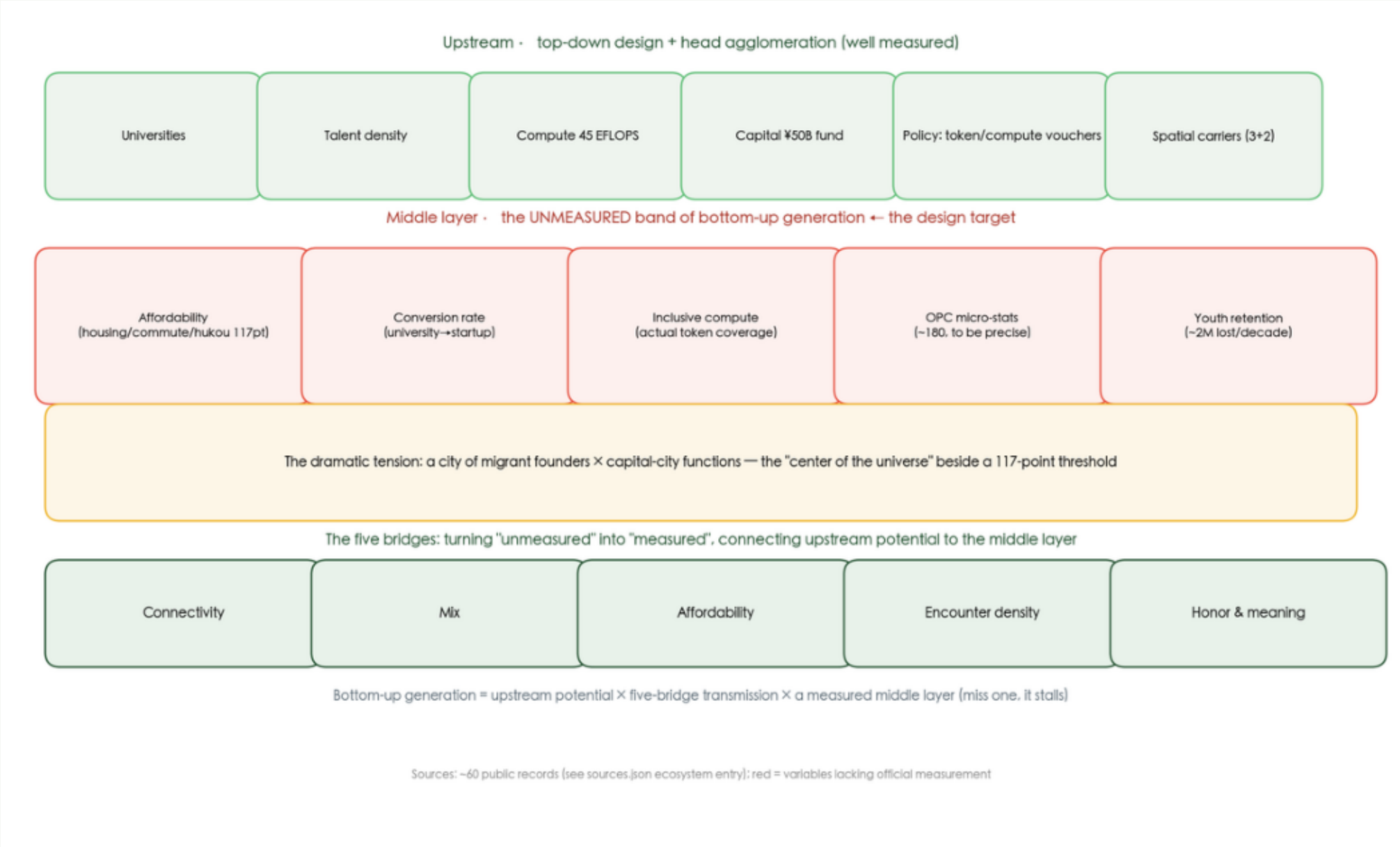
Scenarios & ecosystem: the AI ecology and pilgrimage

Ecosystem map & seven cases · Ten scenario cards, five personas · Three pilgrimage landmarks · Operations & timeline

AI innovation ecosystem map

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- Upstream complete: 1,000+ scientists, 13k developers, 180 OPCs, 32 chips
- Middle layer unmeasured: affordability/conversion/compute → annualized
- Seven cases, one transferable mechanism each (King's Cross / Kendall / Cambridge...)
- Five-party six-force conversion + the annual Phase-Forum report as interface



Operations & the ten-year timeline

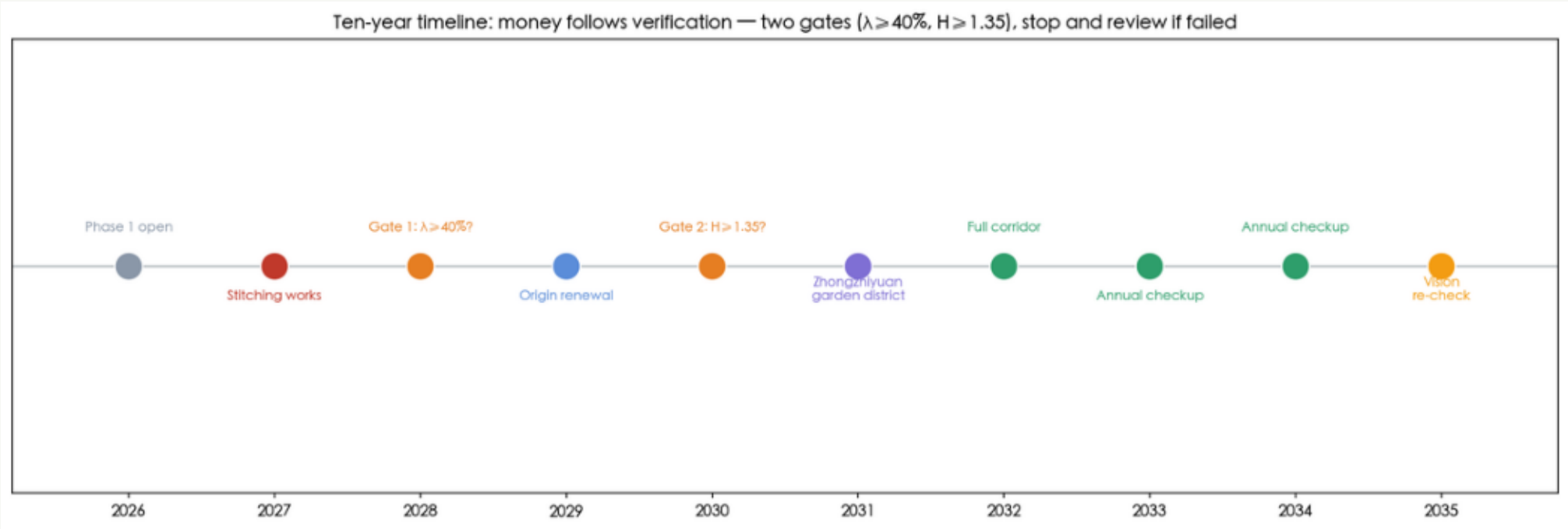
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▪ Annual: dev festival / Phase Forum (λ & H report) / open test seasons

▪ Attraction funnel: OPC contest → order plaza → desks → incubator → conversion

▪ Two acceptance gates: $\lambda \geq 40\%$, $H \geq 1.35$ — or stop and review

▪ Phases: 1–3 stitching+interface / 3–5 Origin / 5–10 Zhongzhiyuan



International benchmark matrix: cases × mechanisms × responses × numbers

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Fig C26 · International benchmark matrix: seven global cases, one transferable mechanism, one of our measured numbers

Benchmark discipline: no copied images or geometry — mechanisms only; each mechanism lands on one of our numbers

Case	Mechanism	Our response	Measured
London · King's Cross	open the paths and squares first, let the plots grow themselves	stitch the seven nodes first, then the plots	integration +43%—+70%
Boston · Kendall Square	the near-campus conversion circle: density within 1 km of MIT	Origin near-campus street: catching those leaving school	Tsinghuayuan integration 93.5%
Stanford Research Park	campus origination: low-density garden research neighbourhood	Zhongzhiyuan garden blocks + the full-stack test band	13,000 developers · 32 chips
Shenzhen · Yuehai Street	the fabric density of high-density startup blocks	Dazhongsi front: a 511 POI/km² station interface	25,476 POIs measured
Tokyo · Shibuya	the renewal paradigm of a content-consumption interface	AI-native commercial front + Emergence Bell (night classes/exhibitions)	10 scenario cards landed
Cambridge Cluster	the symbiotic structure of a university town and an industry belt	the five-party six-force conversion + the annual Phase-Forum report	180 OPCs · 18 vendors adapted
Tel Aviv · Rothschild Blvd	the public life of a startup boulevard	the dev festival + honor wall + spine public space	equity loops 5/5 online

The opportunity: seven mechanisms all hold on one belt — the unique opportunity of Centennial Jing-Zhang: the world's first AI belt designed by urban science and run recursively.

▪ Seven global cases, one transferable mechanism each (no copied images or geometry)

▪ King's Cross × stitch-first / Kendall × conversion circle / Stanford × garden neighbourhoods

▪ Yuehai × dense fabric / Shibuya × consumption / Cambridge × symbiosis / Rothschild × public life

▪ The opportunity: seven mechanisms hold at once on one belt — Jing-Zhang's unique window

Benchmark discipline: mechanisms transferred, numbers computed locally

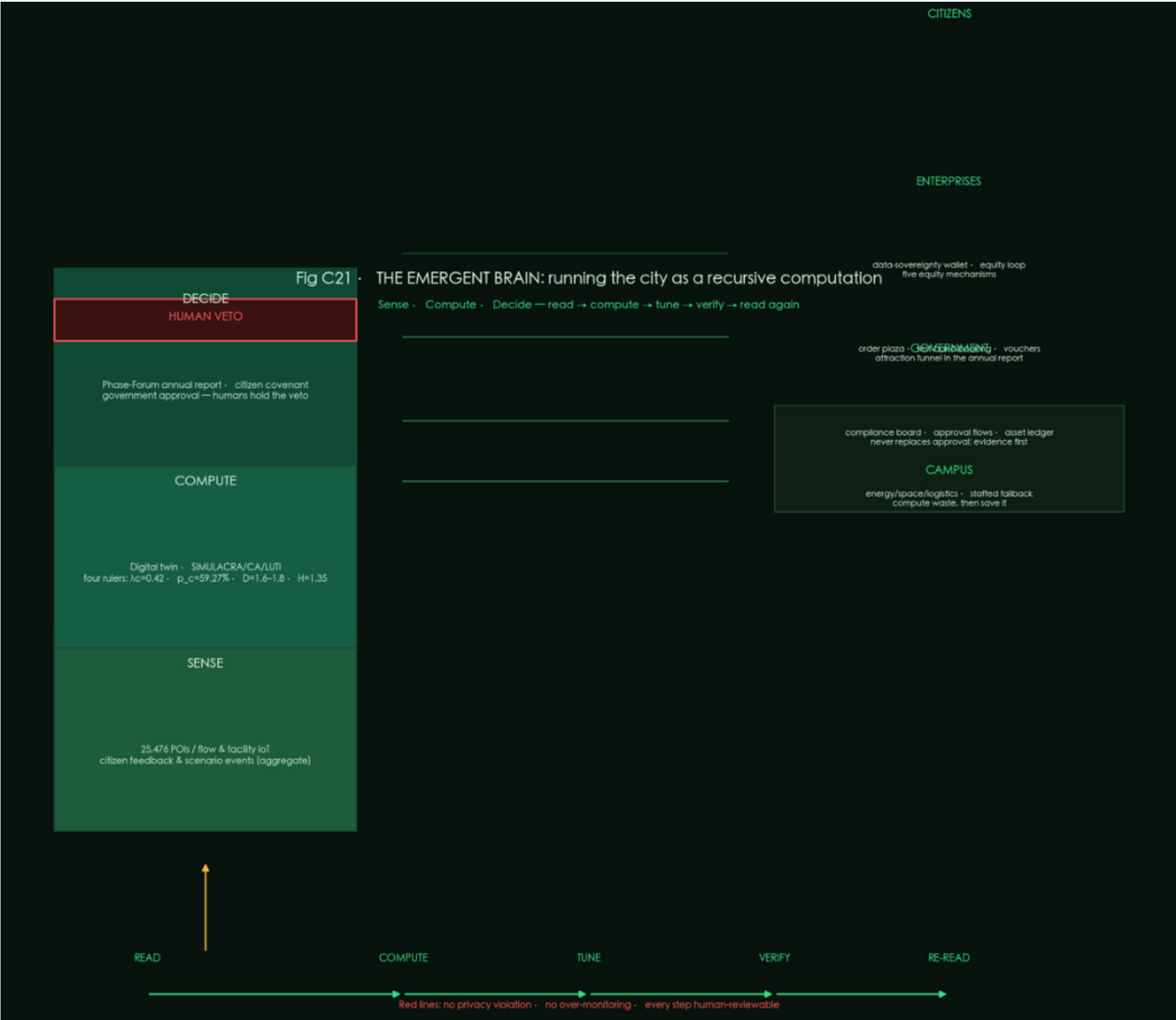
04

System & finance: the Emergent Brain and the investment ledger

Emergent Brain architecture · Phase Control Room · Investment & return · Risk & compliance

📄 The Emergent Brain: the city as a recursive program

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▪ Three layers: sense → compute → decide, with humans always holding the veto

▪ Four terminals: citizens / enterprises / government / campus

▪ The recursive loop is the annual check-up:
read→compute→tune→verify→re-read

▪ Red lines: no privacy violation, no over-monitoring, human-reviewable

Phase Control Room: the belt's annual check-up

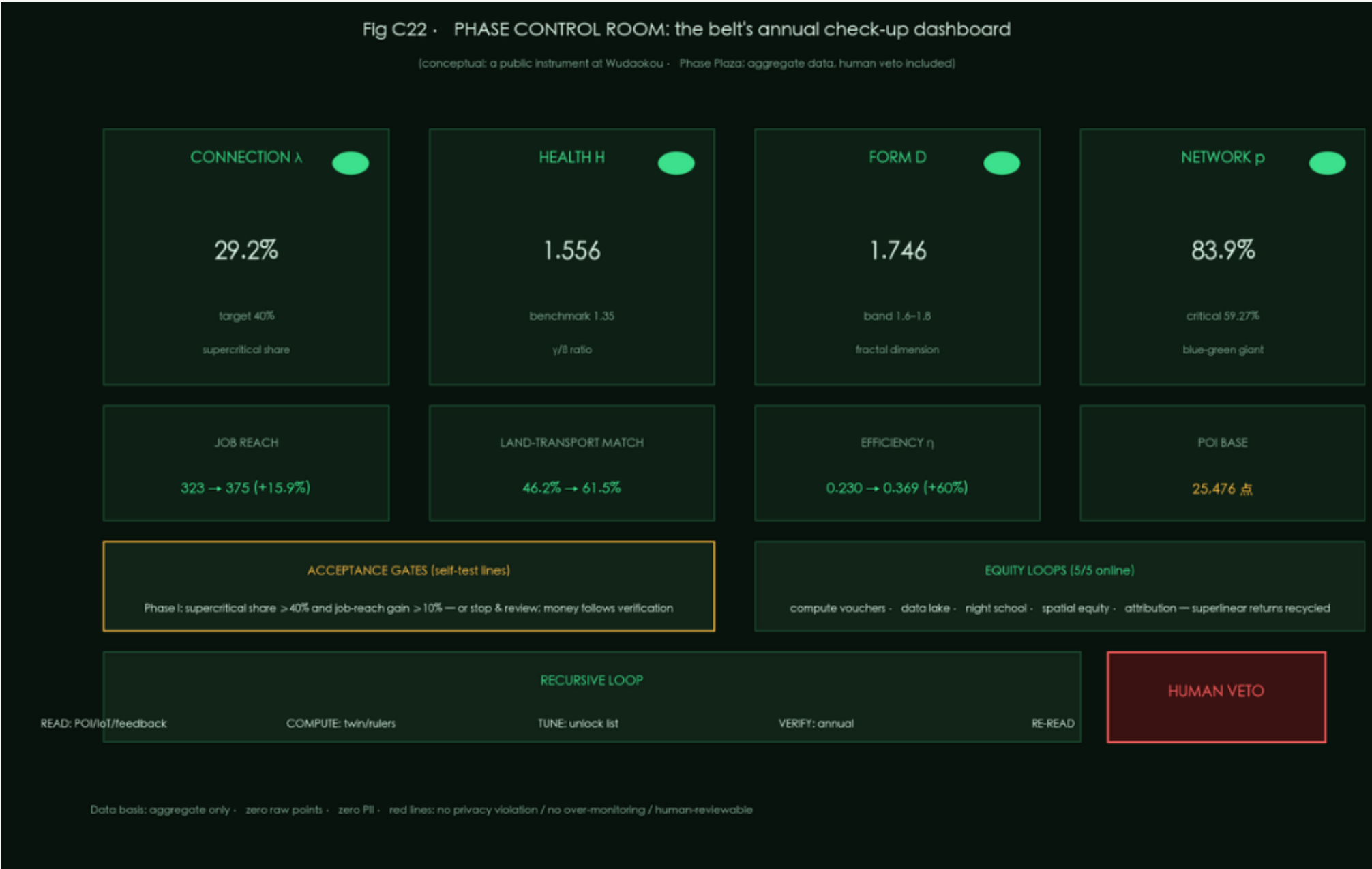
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Four gauges: λ 29.2%→40% target / H 1.556 / D 1.746 / p 83.9%

Live panels: job reach 323→375, match 46.2%→61.5%, η +60%

Physically at Wudaokou · Phase Plaza: a public instrument, not surveillance

Aggregate only, zero raw points, zero PII, human veto



Investment & return (conceptual magnitude)

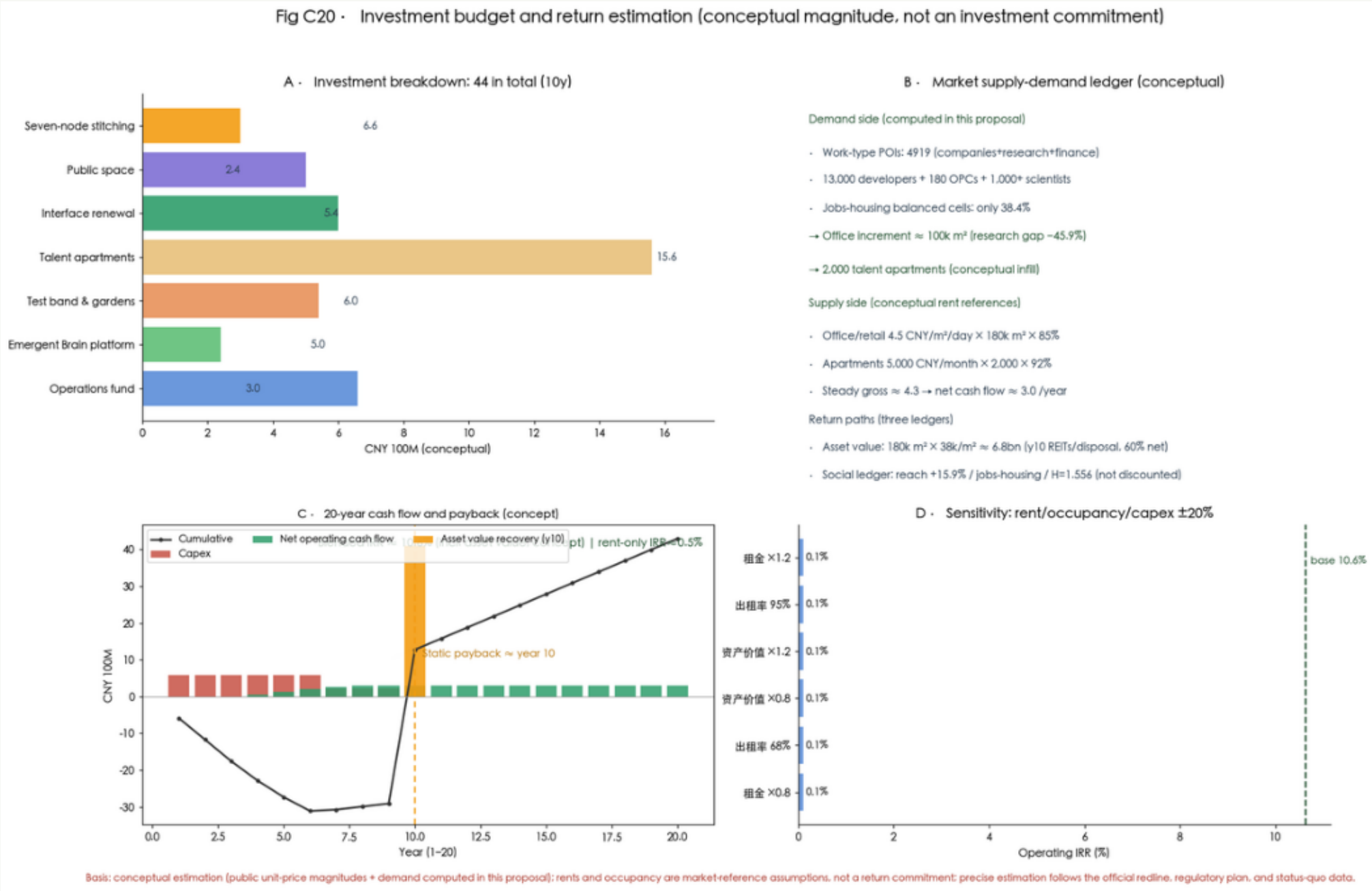
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CNY 4.40bn: stitching 0.66 / space 0.24 / interface 0.54 / apartments 1.56 / test band 0.60 / brain 0.50 / ops 0.30

Three ledgers: rent (IRR≈0.5% alone) + asset (y10 REITs/disposal, blended IRR≈10.6%) + social

Demand: 4,919 work POIs → 100k m² offices; 38.4% balance → 2,000 apartments

Public ledger ≈1.7bn seeks no financial return; market ledger ≈2.7bn runs rent+asset



Conceptual; not an investment commitment; rents & occupancy are replaceable assumptions



05

Conclusion: four sentences, each with numbers

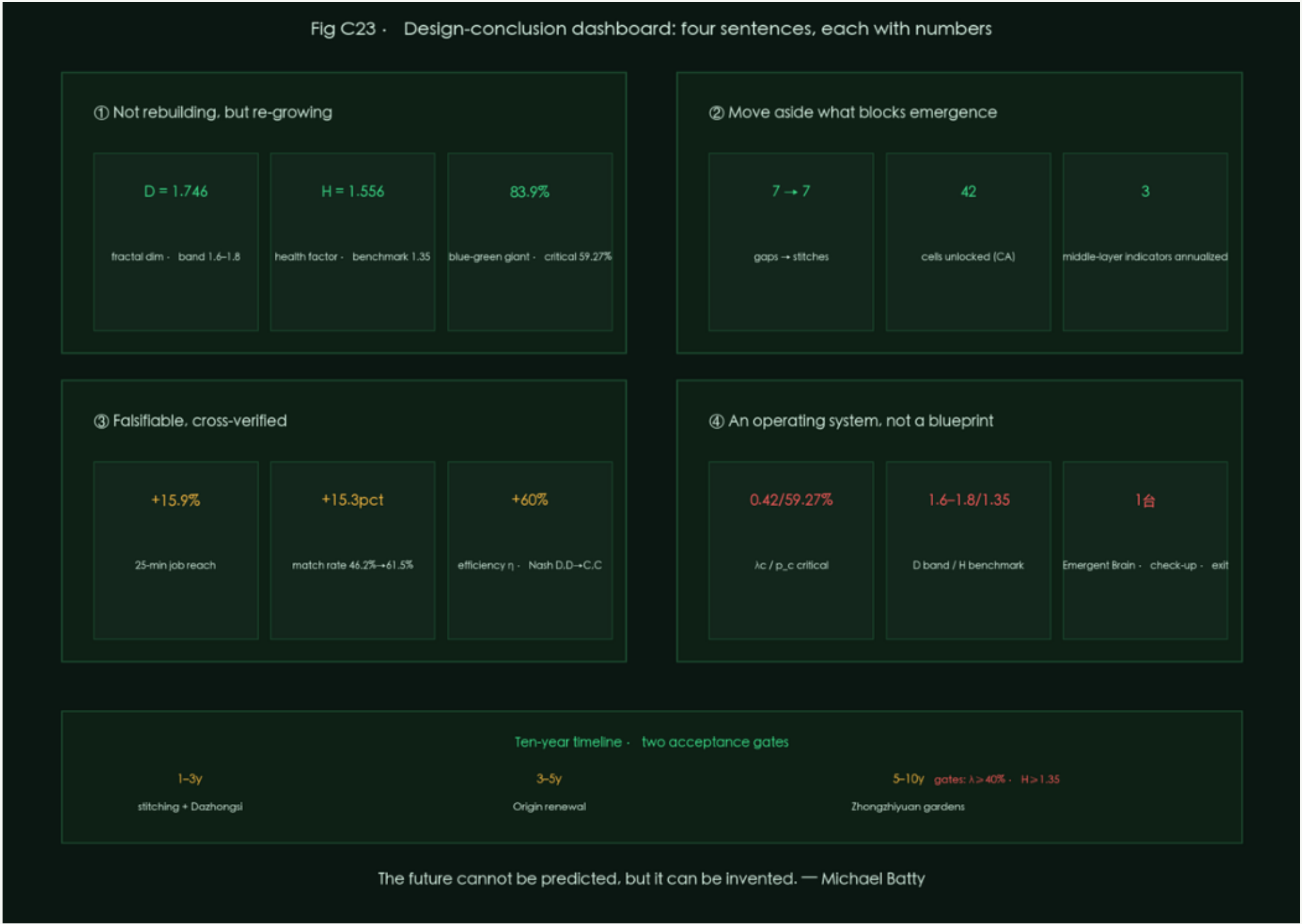
Conclusion dashboard · 1909 switch → 2026 belt

① Re-grow, not rebuild: D=1.746 / H=1.556 / giant 83.9%

② Move aside what blocks: 7 gaps→7 stitches / 42 cells / three annualized

③ Falsifiable: jobs +15.9% / match +15.3pct / η +60%

④ An OS, not a blueprint: four rulers + brain + check-up + exit



Move aside what blocks emergence, and leave the rest for the city to grow itself.

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KUN-SAL Spatial Quant Lab · Xing Xinhua · [GitHub: Anshengdesign](#)

This proposal is an open co-creation suggestion; it does not replace formal planning and constitutes no government-reviewed conclusion; provisional boundaries are dashed only, statutory indicators pending.